Training Dynamics of Trainable Pre-LLM Visual Token Compression: A Failure-Mode Taxonomy

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Abstract. We present a failure-mode taxonomy for the training dynamics of trainable pre-LLM visual token compression in vision-language models at LLaVA-1.5-7B scale; this is an empirical optimisation study, not a benchmark-improvement paper. We build TC-ViTok, a query-conditioned selector that compresses 576 CLIP patch tokens to k=144 before the language model, jointly trained with LLM LoRA adapters, and treat it as a controlled diagnostic substrate. Three coupled optimisation pathologies emerge: (i) *co-adaptation oscillation*, a GAN-style limit cycle between selector and adapter whose period grows from ∼5K to ∼15K steps as the selector learning rate drops 25× in our trainable-pooler family; (ii) *BudgetPredictor collapse*, in which the adaptive budget head pins to its maximum at λ_b=0.03 but remains stable at the single fully measured interior point λ_b=0.10, defining a stable region whose precise boundaries we do not fix; and (iii) *warm-start dependence*, in which every from-scratch joint training run we attempted plateaus 2–10× above the warm-started ceiling regardless of selector variant or LoRA capacity. Each pathology receives a mechanistic explanation, an empirical signature, an architectural or curricular mitigation (a τ-independent additive straight-through estimator correction, a parameter-free informative token merge, a calibrated λ_b, a three-stage curriculum on a converged base), and a falsifiable prediction stated as a reproduction protocol on a different (encoder, LLM, selector) triple. The resulting compressed system loses to a matched same-LoRA no-compression reference by 4.1/3.4/7.4 points on VQAv2/GQA/POPE; we report this cost as part of a controlled diagnostic study. We further find that the learned selector does not surpass the strongest trivial baseline (random, uniform grid, or CLS-attention top-k) on any of five benchmarks at k=144, and the same negative pattern persists across a $k \in [36, 288]$ budget sweep on the adaptive checkpoint, consistent with the warm-start dependence (FM3) we document. We additionally report two empirical properties of the converged selector that we frame as observations, not claims of generality: zeroing the CLIP query embedding at inference yields equal or marginally better accuracy on the Final checkpoint and on its continued training trajectory (we did not test independent seeds for this property), which we interpret as a property of joint optimisation on global-image-distribution training data rather than a defect of the architecture; and a structured per-position selection prior with a localised low-utility region

whose inference-time masking is positive on three checkpoints within a single training trajectory but reverses sign on an independently retrained seed. The durable artefact is the failure-mode taxonomy with falsifiable predictions: a concrete budget for what to expect when attempting trainable pre-LLM compression at this scale.

Keywords: Vision-Language Models · Visual Token Compression · Training Dynamics · Co-Adaptation · Straight-Through Estimator · Query-Conditioned Selection.

1 Introduction

Modern vision-language models (VLMs) such as LLaVA-1.5 [13] attach a large language model (LLM) to a frozen CLIP [18] vision encoder. The encoder maps a 336×336 image to N=576 patch tokens which are prepended to the tokenised question and fed into the LLM. Because LLM attention cost scales quadratically with sequence length, these visual tokens dominate prefill FLOPs, and the budget is set by image resolution rather than by what the question requires.

A natural design point is a *trainable, query-conditioned pre-LLM* compression module that selects $k \ll N$ tokens before the LLM runs. Pre-LLM placement saves the full prefill cost (unlike mid-LLM pruning [3,21]), and query conditioning lets the budget concentrate on relevant regions (unlike query-agnostic approaches [2,12,20]). It is also the design point where joint training fails most interestingly: the selector's output defines the LLM's input distribution, while the LLM's adapters define the gradient signal back to the selector. The resulting closed two-player loop is, structurally, a non-saturating GAN training problem.

This paper is an empirical optimisation-dynamics study of that loop at LLaVA-1.5-7B scale; it is not a benchmark-improvement work. We build TC-ViTok as a controlled instrument for characterising failure modes of trainable pre-LLM compression jointly trained with LLM adapters. We are explicit that the resulting compressed system loses to a matched no-compression LoRA reference (LoRA-576) by 4.1/3.4/7.4 points on VQAv2/GQA/POPE; the compression gap is the setting that makes the analysis possible, and our contribution is the structured account of *why* that gap appears.

Contributions. **(1) A failure-mode taxonomy** for trainable pre-LLM token compression co-trained with LLM adapters. Each pathology receives a mechanism, an empirical signature, and a quantitative characterisation: *co-adaptation oscillation* with period growing as η_{sel} decreases (5K to 15K steps across a 25× learning-rate drop, qualitative trend, not a scaling law); *BudgetPredictor collapse* to $k = k_{\max}$ at λ_b=0.03 with a stable interior point at λ_b=0.10; and *warm-start dependence* in which every from-scratch run plateaus above the warm-started ceiling. Evidence comes from one (encoder, LLM, selector) triple; we present the taxonomy as portable hypotheses, not a universality theorem. **(2) An architectural recipe** that mitigates two of the three pathologies: a τ-independent additive STE correction (Eq. 1); a parameter-free *informative token merge* that

structurally eliminates the co-adaptation loop; and a three-stage curriculum on a converged base. **(3) Two empirical properties** of the converged selector, framed as observations: (a) zeroing the CLIP query at inference is no worse than keeping it on the Final checkpoint and its continued trajectory (independent seeds not tested); and (b) a structured per-position selection prior with a localised low-utility region whose inference-time masking is significant on three checkpoints from one trajectory but reverses sign on an independently retrained seed.

We do not close the $-4.1/-3.4/-7.4$ pp compression cost. The durable artefact is the taxonomy and its predictions; the trained system is the substrate that made them observable.

2 Related Work

Query-agnostic visual token compression. ToMe [2] merges similar tokens via bipartite matching. DynamicViT [19] learns Gumbel-Softmax drop masks distilled from a teacher. EViT [12] ranks tokens by CLS-attention and fuses low-importance ones. PruMerge [20] prunes by CLS rank and merges pruned tokens into nearest neighbours. FastV [3] prunes inside the LLM at layer 2, after visual information has mixed into the text stream. These methods avoid training or rely on strong priors (CLS, layer-2 attention) rather than learn query-aware routing.

Query-aware compression. BLIP-2 [9] uses cross-attention between a fixed bank of learned queries and visual tokens. SparseVLM [21] sparsifies inside the LLM using text-to-visual attention with per-layer recycling. QG-VTC [10] injects the question into the encoder and compresses progressively within it. Focus-LLaVA [22] performs coarse-to-fine instruction-guided selection inside the LLM. None combines (a) explicit per-question conditioning, (b) pre-LLM placement on frozen features, and (c) per-sample adaptive budget; we position TC-ViTok as that combination, primarily so we can study its training dynamics.

Differentiable discrete selection. Gumbel-Softmax [7,14] and straight-through estimators [1] are standard tools for learning subset selectors; AdaViT [15] extends these to per-sample budgets. We adopt Gumbel-Sigmoid for independent binary decisions and contribute a τ-independent additive correction (Section 3.3) that addresses a gradient-vanishing failure mode at low temperature absent from prior work.

3 Method

3.1 Problem Setup and the Co-Adaptation Loop

Given image $\mathbf{I}$ and question $\mathbf{Q}$, we learn a policy $\pi_\phi(\mathbf{I}, \mathbf{Q}) \to \mathcal{S} \subseteq \{1, \ldots, N\}$ with $|\mathcal{S}|{=}k$; the LLM receives only the k selected tokens. Selector parameters ϕ and LLM-adapter parameters θ_{LoRA} are jointly trained against the VQA loss

Table 1: Positioning of TC-ViTok. *Where* is the stage at which compression happens; *Recover* indicates how discarded tokens contribute to the retained representation; *Adapt* k indicates per-sample budget.

Method	**Where**	**Query**	**Recover**	**Adapt** k
ToMe [2]	in-encoder	✗	merge	✗
DynamicViT [19]	in-encoder	✗	✗	✗
EViT [12]	in-encoder	✗	fuse	✗
PruMerge [20]	post-encoder	✗	merge	✗
FastV [3]	in-LLM (L=2)	implicit	✗	✗
SparseVLM [21]	in-LLM	via attn.	recycle	per-layer
QG-VTC [10]	in-encoder	explicit	merge	✗
FocusLLaVA [22]	in-LLM	explicit	coarse-fine	✗
TC-ViTok (ours)	**pre-LLM**	**explicit**	**merge**	**per-sample**

$\mathcal{L}_{\text{VQA}}(\phi, \theta_{\text{LoRA}}) = \mathbb{E}[-\log p_\theta(y \mid \pi_\phi(I, Q))]$. This induces a closed two-player gradient circuit: selector updates shift the LLM input distribution, LoRA updates shift the gradient field the selector sees, both groups respond to a moving target. GAN-style analyses [16,8] predict limit cycles in such circuits, with period growing as either player's update step shrinks. We observe limit cycles empirically (Section 4.1), motivating three design choices: parameter-free recovery (no learnable bridge, Section 3.4), warm-freezing LoRA (break the loop in time, Section 3.5), and adaptive-budget training only on a converged base (Section 4.3).

3.2 Architecture

We use three frozen pretrained encoders: DINOv2-S [17] on a 98×98 crop produces $\mathbf{v}_c \in \mathbb{R}^{49\times384}$ that the selector cross-attends; CLIP ViT-B/32 [18] encodes the question into $\mathbf{q} \in \mathbb{R}^{512}$; and CLIP ViT-L/14 [18] produces the LLM input $\mathbf{v}_f \in \mathbb{R}^{576\times1024}$ (aligned to LLaVA-1.5's projector). No gradient flows into any encoder.

The *Tokenisation Policy Network* (TPN, ≈887K parameters) cross-attends $\mathbf{q}$ over $\mathbf{v}_c$ with four heads to produce a context vector $\text{ctx} \in \mathbb{R}^{384}$ and head-averaged spatial weights $\boldsymbol{\alpha} \in \mathbb{R}^{49}$. A small score MLP on $[\text{ctx}; \boldsymbol{\alpha}]$ emits 49 coarse scores, bilinearly upsampled to a 24×24 grid and *instance-normalised* to $\mu{=}0, \sigma{=}1$ per sample. Instance normalisation is necessary: without it Adam pushes scores to ±100 within ∼1K steps and Gumbel noise becomes negligible; `tanh` (saturation) and batch norm (single-sample inference) failed analogously.

3.3 Gumbel-Sigmoid STE with τ-Independent Correction

Token selection is a discrete top-k operation. We use Gumbel-Sigmoid STE for independent binary subset selection over 576 positions. With $\mathbf{g} \sim \text{Gumbel}(0, 1)$

and τ annealed linearly $1.0 \to 0.3$: $\mathbf{n}=\mathbf{s}_N+\mathbf{g}$, $\mathbf{m}_{\text{soft}}=\sigma(\mathbf{n}/\tau)$, idx=topk$(\mathbf{n}, k)$, and $\mathbf{m}_{\text{STE}}=\mathbf{m}_{\text{hard}} - \mathbf{m}_{\text{soft}}.\text{detach}() + \mathbf{m}_{\text{soft}}$. The gradient flows through $\mathbf{m}_{\text{soft}}$.

At low τ, $\mathbf{m}_{\text{soft}}$ saturates and $\partial \mathbf{m}_{\text{soft}}/\partial \mathbf{s}_N \approx 0$ for $|\mathbf{s}_N[i]| > 1$, killing the gradient to the selector. Letting $\mathbf{c}=\text{gather}(\mathbf{s}_N, \text{idx})$, we apply

$$\mathbf{v}_{\text{sel}} \leftarrow \mathbf{v}_{\text{sel}} \odot \big(1 + \mathbf{c} - \mathbf{c}.\text{detach}()\big). \tag{1}$$

Forward: $\mathbf{c}-\mathbf{c}.\text{detach}()=\mathbf{0}$, so $\mathbf{v}_{\text{sel}}$ is unchanged. Backward: for selected position j, $\partial \mathcal{L}_{\text{VQA}}/\partial \mathbf{s}_N[j]=\sum_d (\partial \mathcal{L}_{\text{VQA}}/\partial \mathbf{v}_{\text{sel}}[j,d])\, \mathbf{v}_{\text{sel}}[j,d]$, a dot product between the VQA gradient and the visual features at j that is non-zero at any τ. Without Eq. 1, TPN gradients die around step 15K in otherwise identical runs.

3.4 Informative Token Merge (Parameter-Free)

Hard dropping of the $N-k=432$ unselected tokens discards their context. We route them into selected tokens via cosine similarity weighted by TPN scores, with normalisation $\tilde{A}_{ij}=w_i A_{ij} / \sum_{i'} w_{i'} A_{i'j}$ where A_{ij} is a row-softmax over cosine similarity at temperature τ_m and w_i is a column-softmax over unselected-token scores:

$$\mathbf{v}_{\text{merged}} = \mathbf{v}_{\text{sel}} + \beta\, \tilde{A}^{\top} \mathbf{v}_{\text{unsel}}, \qquad \beta=0.3,\ \tau_m=0.1. \tag{2}$$

The merge has *zero new parameters*, eliminates co-adaptation risk with LoRA by construction, and preserves the token count k between training and evaluation. A previous family of eight *trainable* pooler variants displayed the closed-loop oscillation diagnosed in Section 4.1; the parameter-free merge structurally severs one edge of the gradient circuit and eliminates those failures.

3.5 Adaptive Budget and Training Curriculum

A small ≈25K-parameter *BudgetPredictor* head maps $[\text{ctx}; \bar{h}(\mathbf{s}_N)]$ to a per-sample budget fraction $k_{\text{frac}} \in [36/576, 288/576]$; the adaptive variant adds a penalty $\mathcal{L}_b=\lambda_b\, \mathbb{E}[k_{\text{eff}}/N]$ to the objective. The collapse modes of this head are characterised in Section 4.2.

All TC-ViTok runs use a three-stage curriculum: **Phase 0** trains the TPN alone (LLM frozen); **Phase 1** closes the joint loop with LoRA warm-frozen for the first 3K steps; **Phase 1 Adaptive** adds the BudgetPredictor on top of a converged Phase 1 checkpoint. We identify this curriculum by elimination: from-scratch joint training and re-orderings of the three stages did not converge in our experiments.

4 A Failure-Mode Taxonomy for Trainable Pre-LLM Compression

We document three coupled pathologies. Each receives a mechanistic explanation, an empirical signature, and a remedy where one exists. **Scope of evidence.** The

controlled evidence comes primarily from one (encoder, LLM, selector) triple: CLIP ViT-L/14 features, LLaVA-1.5-7B with LoRA adapters, and the TPN selector of Section 3.2. We add a Qwen2-VL-7B stress test for FM1, but do not claim taxonomy-level universality. Each pathology is stated with a falsifiable prediction and a reproduction protocol (Section 4.4); whether the predictions hold broadly is an open empirical question.

4.1 FM1: Co-Adaptation Oscillation

Prediction. We predict that other architectures sharing this gradient-circuit structure — a trainable parameter bridge between a learnable pre-LLM selector and adapter-trained LLM weights — will exhibit limit-cycle behaviour in joint training, with cycle period increasing monotonically as either group's learning rate decreases. We state this as a prediction, not a theorem: our evidence is eight selector variants in one (encoder, LLM, selector) triple, and the functional form of the period–LR relationship is not constrained by our two-point sweep (see Signature).

Mechanism. The two-player circuit of Section 3.1 is structurally identical to a non-saturating GAN training problem. When the selector contains a trainable bridge to the LLM, both groups update at finite rate and chase each other; GAN convergence analyses [16,8] predict limit cycles whose period grows as either player's update step shrinks, with the joint dependence determined by the eigenvalues of the coupled gradient field rather than by a closed inverse-LR form.

Signature. Across eight architecturally distinct trainable-pooler variants we observed oscillatory validation-VQA trajectories. Figure 1 shows three representative runs. At $\eta_{\text{sel}}=5\times10^{-4}$ with η_{LoRA} held at 2×10^{-5} (selector 25× faster than LoRA), period $\sim$ 5K steps; at $\eta_{\text{sel}}=2\times10^{-5}$ (both groups matched at the same rate), period $\sim$ 15K steps; freezing the pooler ($\eta_{\text{sel}}=0$) eliminated oscillation but produced fixed noise tokens the LLM could not use. The three-fold period increase under a twenty-five-fold η_{sel} reduction is consistent with the bottleneck shifting from LoRA-limited to jointly-limited dynamics rather than with a simple inverse-η_{sel} relationship; two finite-period measurements do not constrain the functional form. We report the period–LR relationship as a qualitative trend supporting FM1's existence, not its scaling.

Cross-architecture stress test. We reproduce the trainable-bridge optimisation on Qwen2-VL-7B with LoRA and a cross-attention pooler at two selector learning rates ($\eta_{\text{sel}}=5\times10^{-4}$ and 2×10^{-5}), 12K steps each. Both runs show clearly non-monotonic validation loss (peak-to-trough $\Delta\approx2.1$ at the higher rate, $\Delta\approx6.9$ at the lower rate), showing that trainable-bridge instability is not LLaVA-specific. Exact-match accuracy on a 500-sample GQA subset is too sparse (mostly zero, max 0.012) to fit a period, so we do *not* extract a cross-architecture period–LR relationship; this stress test supports trainable-bridge instability on a second VLM stack, not FM1's period scaling.

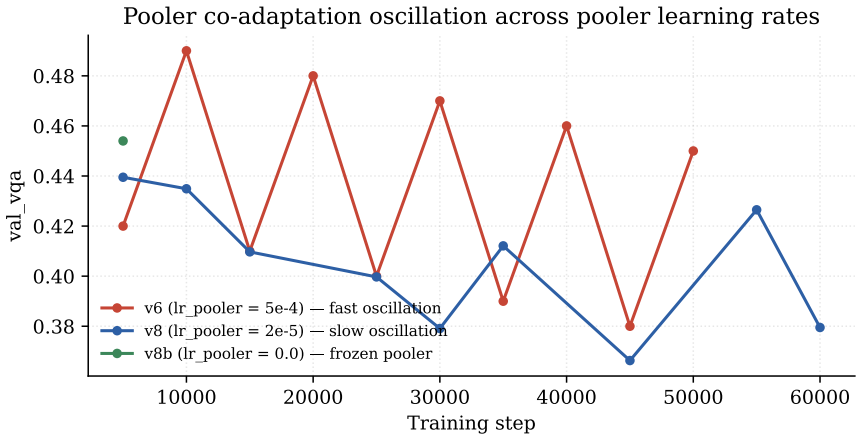


Fig. 1: Co-adaptation oscillation across selector learning rates. *Top:* $\eta_{\text{sel}}=5\times10^{-4}$, period $\approx$ 5K steps. *Middle:* $\eta_{\text{sel}}=2\times10^{-5}$, period $\approx$ 15K steps. *Bottom:* $\eta_{\text{sel}}=0$ (frozen pooler), no oscillation but fixed noise tokens the LLM cannot use.

Remedy. Replace the trainable bridge with the parameter-free informative token merge (Section 3.4). With no learnable connector, one edge of the gradient circuit is structurally absent and the loop cannot close. All eight trainable-pooler variants oscillated; the parameter-free TC-ViTok Final does not.

4.2 FM2: BudgetPredictor Collapse

Prediction. Any adaptive-k head supervised by a monotone-in-k task gradient and a budget penalty $\mathcal{L}_b = \lambda_b \mathbb{E}[k_{\text{eff}}/N]$ *has a collapse boundary below which* $k_{\text{eff}} \to k_{\max}$ *uniformly, and an unbounded upper boundary above which task accuracy degrades. The existence of a non-empty stable window is the prediction; its location depends on architecture.*

Mechanism. The budget head receives two opposing gradient signals: $\mathcal{L}_{\text{VQA}}$ pulls k_{eff} *up* (more tokens generally ease VQA), $\mathcal{L}_b$ pulls it *down*. When the former dominates, the predictor converges to the trivial saturating policy $k_{\text{frac}} \to k_{\max}/N$ uniformly and the adaptive variant degenerates to fixed-$k_{\max}$ with a dead-weight head.

Signature. Fine-tuning the adaptive variant from a warm-started base, we observe a sharp transition (Table 2). At λ_b=0.03, k_{eff} pins to 287.97/288 within 10K steps (collapsed) and validation loss regresses; at λ_b=0.10, $k_{\text{eff}} \approx 271$ is stable for the full 20K-step horizon. Two larger values were not run to completion but projected behaviour places the upper boundary near λ_b=0.15. We report the empirically stable window as $\lambda_b \in [0.08, 0.15]$ with one measured interior point.

Table 2: BudgetPredictor outcome during fine-tuning of the warm-started checkpoint. Collapse: $k_{\text{eff}} > 0.99\, k_{\max}$ for $\geq$ 2K consecutive steps. The λ_b=0.10 point is the only one run to completion.

λ_b	k_{eff} (steady)	Outcome	Val-VQA
0.03	287.97/288	collapsed (by 10K)	0.3898 (regressed)
0.10	∼271 / 288	stable	0.3177
0.15†	∼255 / 288	stable (projected)	—
0.30†	∼200 / 288	accuracy-dominant pressure	—

† Boundary values not run to completion.

Remedy. Operate inside $\lambda_b \in [0.08, 0.15]$ and apply the budget head only on a converged base. The failure mode is not specific to our architecture: it arises whenever a budget penalty is the only counterweight to an unbounded task gradient acting through a monotone-in-k soft mask — a recurring pattern in adaptive compression [15,19].

4.3 FM3: Warm-Start Dependence

Prediction. Joint random-initialisation training of a high-capacity selector and LLM adapters at ≥7B-LLM scale, supervised by end-task loss alone, plateaus above any warm-started equivalent unless an additional supervisory signal (distillation, oracle attention, or a strong prior on the selector) is introduced. This prediction is the weakest of the three because the existence of any one effective supervisory augmentation would falsify it; we invite that test.

Mechanism. A randomly-initialised TPN gives the LLM a near-uniform selection distribution; LoRA faces a flat task landscape and adapts in directions that are information-poor for the converged regime. The selector then receives gradient signal back through a LoRA trained on near-uniform input. Neither parameter group offers the other useful signal until at least one reaches a non-trivial regime. The co-adaptation loop is not oscillating here; it is *stuck*.

Signature. Across four independent large-scale runs spanning two selector variants, three LoRA capacities, and four learning-rate regimes, every from-scratch joint training plateaued well above the warm-started ceiling (Table 3). The only configuration that approached the ceiling started from a converged base.

Remedy. We do not defeat this failure mode. The three-stage curriculum treats warm-start as a necessary condition: adaptive-budget training is applied *only* on top of a converged Phase 1 base.

Table 3: Warm-start vs. from-scratch outcomes on the joint TPN+LoRA+projector system. Val-VQA is NTP cross-entropy on VQAv2 val (lower is better).

Initialisation	**TPN**	**LoRA** r/α	**Best Val-VQA**	**Steps**
warm-start (P1-Final ckpt)	v1	64 / 128	**0.3177**	40K
from scratch	v1	64 / 128	0.4915 (oscillating)	48K
from scratch	v3	32 / 64	0.4380 $\rightarrow$ 0.39 (regressing)	40K
from scratch	v3	32 / 64†	0.3334 (still descending)	50K

† Image-only TPN; reached the same ceiling but on a single seed and only by step 50K.

4.4 Testing the Taxonomy on Other Architectures

Each failure mode carries a reproduction protocol on a different (encoder, LLM, selector) triple, intended to make the predictions in Sections 4.1–4.3 testable by independent groups without re-deriving the analysis.

Reproduction protocols. FM1: instantiate a trainable bridge (learned pooler, cross-attention summariser, or auxiliary projector) between a frozen visual encoder and an adapter-trained LLM, sweep selector learning rates over at least one decade, and test whether oscillation period grows as update rates shrink; replacing the bridge with a parameter-free recovery operator should remove the cycle. FM2: add an adaptive-budget head with penalty $\lambda_b \, \mathbb{E}[k_{\text{eff}}/N]$, sweep λ_b from 0.03 to 0.30, and test for a collapse boundary below a non-empty stable region. FM3: train from random initialisation; the prediction is a plateau 2–10$\times$ above the warm-started ceiling unless an additional supervisory signal, such as distillation or an oracle selector prior, falsifies it.

5 Experiments

5.1 Setup

Hardware and data. All experiments run on 2$\times$ RTX 6000 Ada (49 GB each) with PyTorch DDP, BF16 mixed precision, and effective batch size 128. We train on VQAv2 [5] and GQA [6] and evaluate on VQAv2 val2014, GQA val_balanced, POPE [11], MMBench English dev, MME, and TextVQA val.

The missing control. Comparing a LoRA fine-tuned compressive system against an off-the-shelf uncompressed baseline conflates compression with LoRA effects. We introduce LoRA-576: the same Phase-1-Final LoRA adapter and multimodal projector evaluated with all 576 tokens forwarded to the LLM, byte-identical to TC-ViTok Final in rank, alpha, dropout, targets, SFT data, optimiser, and checkpoint step. The architectural comparison is TC-ViTok Final versus LoRA-576, both at $\sim$60 GPU-h compute; published query-agnostic methods are not compute-matched.

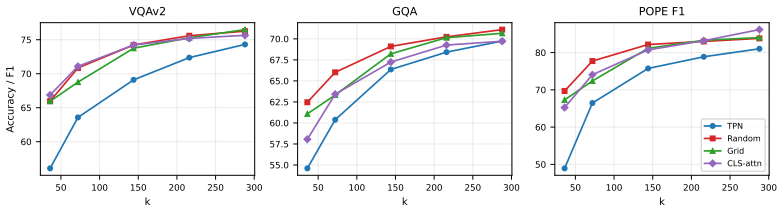


Fig. 2: Token-budget sweep on the Phase 1 Adaptive checkpoint. The learned TPN remains below the strongest trivial selector at every measured k on VQAv2, GQA, and POPE.

5.2 Main Results

TC-ViTok Final costs $-4.1/-3.4/-7.4$ pp on VQAv2/GQA/POPE against LoRA-576 (Table 4). The same pattern persists on an independently retrained seed (Table 5): the gap is not a seed-42 artefact. Against published query-agnostic methods, TC-ViTok underperforms by $\sim$5 points; those methods use the full publicly released LLaVA-1.5 SFT mix that LoRA-576 also lacks, so the compute-matched architectural cost is the gap to LoRA-576.

Trivial-selector baselines. We evaluate three trivial selectors at the same k=144, same LoRA adapter, and same hard-drop path: uniform random selection, uniform grid (every 4th patch), and CLS-attention top-k on CLIP's last layer (Table 4). *The learned TPN does not surpass the strongest trivial baseline on any of VQAv2, GQA, POPE, MMBench, or TextVQA.* On VQAv2 and TextVQA the TPN loses to all three trivial selectors; on GQA and POPE the TPN loses to random drop and uniform grid while marginally exceeding CLS-attention top-k; on MMBench the TPN beats random and uniform but loses to CLS-attention top-k. The TextVQA gap is the most striking instance: CLS-attention top-k on the same 24×24 patch grid exceeds the TPN by 10.8 pp, indicating that the joint-training objective did not refine the selector toward text-localisation regions even though that prior is present in CLIP's last-layer attention. The overall pattern is consistent with FM3 (warm-start dependence): the ceiling we approach is bounded by what trivial selection already extracts from this LoRA adapter, and joint end-task supervision through the LLM does not push the learned selector beyond that ceiling.

Scaling with token budget. The negative finding extends across budgets. On the Phase 1 Adaptive checkpoint, evaluating the same four selectors at $k \in \{36, 72, 144, 216, 288\}$ shows that the learned TPN does not beat the strongest trivial baseline at any measured k on VQAv2, GQA, or POPE (Fig. 2). The gap narrows as k increases but does not reverse, supporting the interpretation that joint end-task supervision through the LLM fails to extract value over trivial selection in this regime rather than only at the headline k=144.

Table 4: Main results on LLaVA-1.5-7B. Headline comparison is TC-ViTok Final vs. LoRA-576. All TC-ViTok rows are single-seed evaluations; binomial standard errors at the listed sample sizes are $\sigma \approx 0.16$ pp for VQAv2 testdev, $\sim$0.22 pp for GQA, and $\sim$0.50 pp for the 5K POPE subset.

Method	**Tok**	**VQAv2**	**GQA**	**POPE F1**	**MMB**	**TextVQA**
Published (LLaVA-1.5 SFT, no extra LoRA):						
LLaVA-1.5-7B (full) [13]	576	78.5	61.0	84.4	64.3	58.2
Same-LoRA reference and ours (this work, $\leq$80K steps):						
LoRA-576 (control)	576	76.38	70.55	86.46	61.7	38.4
TC-ViTok v3 (hard drop)	144	72.47	59.87	78.50	—	—
TC-ViTok Adaptive	36–288	69.43	65.04	75.63	—	—
TC-ViTok Final	**144**	**72.27**	**67.12**	**79.09**	**62.0**	**25.8**
merge disabled at eval	144	72.27	67.12	79.09	—	—
zero CLIP query at eval	144	72.67	67.45	79.37	—	—
Trivial-selector baselines (same LoRA, k=144, hard drop):						
Random drop	144	74.05	70.40	83.92	61.24	29.63
Uniform grid	144	74.65	69.26	82.78	61.93	29.61
CLS-attention top-k	144	74.91	68.16	78.94	62.05	36.57
Published query-agnostic methods (different SFT mix and compute):						
FastV [3]	144	77.1	—	—	—	—
LLaVA-PruMerge [20]	144	76.8	—	—	—	—
SparseVLM [21]	$\sim$152	78.2	—	—	—	—

5.3 Properties of the Converged Selector

Interpreting query-agnostic convergence. The observation that zeroing the CLIP query at inference does not degrade accuracy is a property of the *optimisation landscape at convergence*, not a verdict on the architecture. The CLIP query channel is causally active during training (from-scratch image-only training is slower than from-scratch query-conditioned training, Table 3), consistent with the question signal stabilising the selector against early entropy collapse. At convergence, the learned spatial-attention bias approximately suffices and the query channel becomes redundant. We present query-agnostic convergence as an empirical property of joint pre-LLM selector–adapter optimisation, not as a defect of TC-ViTok, and we hypothesise that it recurs in pre-LLM trainable selectors trained under similar conditions (global-image-distribution datasets such as VQAv2 and GQA, without explicit grounding-task data such as RefCOCO and ChartQA).

Measured behaviour. Replacing the CLIP question embedding with a zero vector at inference yields $+0.40/+0.33/+0.28$ pp on VQAv2/GQA/POPE on the Final checkpoint (Table 4). Repeating on a continuation of the same training trajectory (Adaptive v2, 5K samples) gives baseline 69.10/66.36/75.74 vs. zero-

Table 5: Two-seed replication of the matched compression gap. Both rows are 5K-sample evaluations after 80K-step Phase 1 training with identical configuration and different seeds.

Seed	Model	VQAv2	GQA	POPE F1
42	TC-ViTok Final	72.27	67.12	79.09
42	LoRA-576	76.38	70.55	86.46
7	TC-ViTok Final	72.34	67.58	78.87
7	LoRA-576	76.65	71.36	86.78

Table 6: Inference-time merge-weight sweep on the Phase 1 Final checkpoint. Variations are at the level of subset noise.

β	VQAv2	GQA	POPE F1
0.3	72.96	68.98	79.19
0.5	72.76	68.80	79.12
0.7	73.10	68.92	79.04
1.0	72.89	68.86	79.03

query 69.66/66.20/76.15, a pattern of no material degradation (GQA slightly negative). We did not test the zero-query intervention on an independent seed, so this finding is trajectory-internal.

Token merge: architectural, not benchmark-improving. The merge-disabled ablation (Table 4) is byte-identical to the merged result, and a four-point inference-time sweep of the merge weight β from 0.3 to 1.0 (Table 6) yields all points within 0.34 pp on VQAv2, 0.18 pp on GQA, and 0.16 pp on POPE on a 5K subset. The merge's contribution is *architectural*: it eliminates the trainable bridge that produces FM1 (Section 4.1), not downstream accuracy.

Structured per-position selection prior. Aggregating selection statistics over 500 GQA val_balanced samples (Fig. 3), per-position selection frequency ranges 0–0.7 with a long near-zero tail, and per-image mean normalised score is approximately uniform except for a localised upper-centre region (rows 0–4, cols 6–11) that is systematically deprioritised. We use inference-time masking of this region as a probe. The probe gives +2.2 pp exact-match VQA-score on Final (95% CI $[+0.4, +4.4]$, p=0.011 vs. 10 size-matched random rectangles) and a similar positive sign on two same-trajectory descendants (Adaptive v2 +2.4, v3 +2.2). On an independently retrained seed-7 checkpoint the probe *reverses sign*: −1.0 pp, CI $[-2.8, +0.8]$ (consistent with zero). Cross-dataset overlap with VQAv2-derived low-score positions is narrow (Jaccard 0.067). We present the deprioritised region as trajectory-internal: reproducible within one training run but not across seeds. We claim no masking remedy and do not isolate the underlying cause (attention sink, content prior, or LoRA artefact).

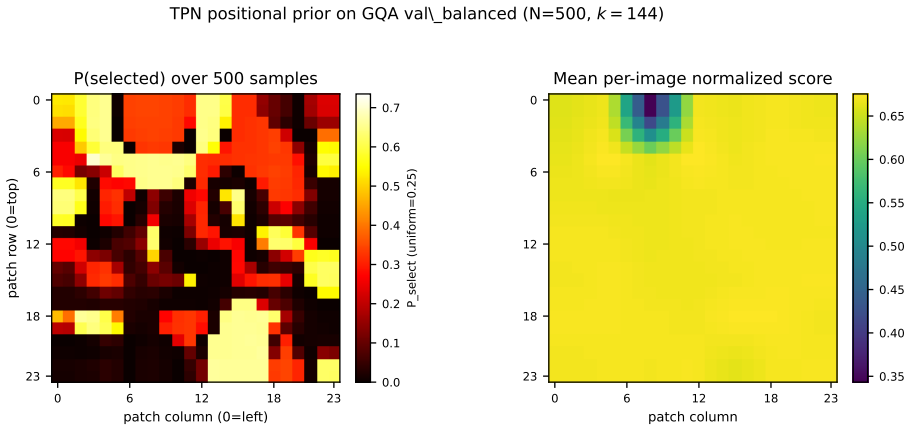


Fig. 3: Aggregate selection diagnostics on 500 GQA val_balanced samples (Phase 1 Final). *Left:* per-position selection frequency P_{select} shows range 0–0.7 with mild upper-region preference. *Right:* per-image min-max normalised score is approximately uniform except for a localised upper-centre region (rows 0–4, cols 6–11) that the TPN systematically deprioritises regardless of content.

Adaptive variant as a design exploration. Phase 1 Adaptive at λ_b=0.10 converges to mean $k_{\text{eff}} \approx 271$ inside the stable region of Section 4.2, confirming that the BudgetPredictor head can be trained in the predicted stable regime. Because the adaptive variant underperforms Phase 1 Final and is confounded by shorter training, LoRA target mismatch, and variable-k optimisation, we treat it as an FM2 existence proof rather than an end-to-end accuracy contribution.

6 Discussion

Why TC-ViTok loses to its matched baseline. Three observations explain the $-4.1/-3.4/-7.4$-point gap. First, the selector's gradient signal arrives via a 32-layer LLM and a frozen projector; in contrast to systems that distil from a strong teacher [19], our selector is supervised only by the end-task cross-entropy, a distal signal. Second, the merge β sweep (Table 6) shows LoRA absorbs the merge's information flow at every tested β; the LLM adapts to the post-merge representation, shadowing the merge's accuracy contribution. Third, pre-LLM compression cannot exploit the early-layer attention mixing that mid-LLM methods like FastV [3] rely on: information lost before layer 0 is lost for all 32 layers, and the selector must anticipate the LLM's needs without ever seeing an LLM activation. The TextVQA gap (TC-ViTok Final 25.8 vs. LoRA-576 38.4 on a 5K subset) is the largest per-benchmark gap to the no-compression baseline. CLS-attention top-k on the same 24×24 patch grid reaches 36.6 at k=144 (Table 4), so the gap is not a coarseness ceiling of the grid itself: it is a property of the spatial distribution the TPN learned. We flag pre-LLM compression at

k=144 as most damaging in text-heavy regimes, where end-task joint training does not push the selector toward the text-localisation prior that CLIP's own CLS-attention already encodes.

Limitations. (1) The headline rows are single-seed; one independently retrained seed (Table 5) replicates the compression gap but *does not* replicate the dead-zone masking effect. (2) Primary taxonomy evidence comes from one (encoder, LLM, selector) triple; a Qwen2-VL-7B stress test with a cross-attention pooler (Section 4.1) shows non-monotonic validation loss on a second VLM stack but does not reproduce the period–LR ordering on its sparse exact-match signal. (3) The $-4.1/-3.4/-7.4$-point gap to LoRA-576 is not closed, and the -12.6-point TextVQA gap is the worst per-benchmark instance. (4) The dead-zone masking effect is trajectory-internal: positive on three descendants of the same Phase-1 trajectory, negative on an independent seed, and consistent with no robust effect. (5) The FM2 stable region is supported by one measured interior point; the two boundary values were not run to completion. (6) The $T_{\mathrm{osc}}(\eta_{\mathrm{sel}})$ relation rests on two finite-period measurements plus a frozen-pooler control — a qualitative trend, not a scaling law. (7) At k=144 the learned TPN is beaten by at least one trivial baseline (random, uniform grid, or CLS-attention top-k) on every one of five benchmarks (Table 4), and the negative pattern persists across $k \in \{8, 18, 24, 36, 72, 144, 216, 288\}$ on the adaptive checkpoint (Fig. 2 shows the in-distribution range); the contributions are the failure-mode taxonomy and its mitigations, not a positive learned-selection recipe.

Future work. Natural extensions are grounding-style evaluations, register-token CLIP encoders [4], and distillation from a stronger teacher to address the distal-supervision pathway above.

7 Conclusion

We treated a trainable, query-conditioned pre-LLM visual token selector as an instrument for studying joint selector–adapter optimisation at LLaVA-1.5-7B scale. TC-ViTok does not beat its matched no-compression baseline; its scientific value lies in what it makes observable. Three coupled pathologies emerge: co-adaptation oscillation, BudgetPredictor collapse, and warm-start dependence. The τ-independent STE correction, the parameter-free informative token merge, and the three-stage curriculum mitigate two of the three; the third is reported as a necessary condition. At convergence, the selector is empirically close to query-agnostic and exhibits a structured per-position low-utility allocation that does not replicate across independent seeds. It also does not surpass the strongest trivial baseline (random, uniform grid, or CLS-attention top-k) on any of five benchmarks at k=144 or across a $k \in [36, 288]$ sweep; the taxonomy explains why end-task supervision through the LLM does not push joint training beyond what trivial selection already achieves. The durable artefact is the taxonomy and its falsifiable predictions; the trained system is the substrate that made them observable.

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